

Tunable Rocky Planet Water Evolution Model: Specification v0.1

Navigator's Log R&D. Draft started 3 October 2026. Status: DRAFT. Rules are not frozen; no test runs have been made.

Project status and stopping point

Checkpoint 20, 4 October 2026. Updated at every stopping point so work can resume cleanly. [CL]

Current stage: 2 of 6, equations and coupling table (in progress). Working document: [Stage 2, Process Inventory and Coupling Table v0.1](#). Stages [CL]: 1 specification; 2 equations and coupling table; 3 parameter sheet in Google Sheets; 4 Python reference model; 5 freeze, hash and first test run; 6 visual build and deployment.

- Done: stage 1 closed; all [CL] decisions up to Checkpoint 5 confirmed by CBH; evidence catalog verified (references 1 to 92, 97 and 98, with 93 to 96 as V*); all tests and tolerances approved; Stage 2 companion document started with the process inventory and coupling table; accretion run design part 1 done: a collision-outcome model with hit-and-run and fragmentation found, verified and smoke-tested (refs 99 to 105); D11 decided by CBH (option C, two iron scenarios); accretion run design part 2 drafted (hosting, benchmark B0, decision D12).
- Next: D12 decided [CBH]: matched comparison P-D12; P-D12 margins approved by CBH (Stage 2 document); repository files drafted and tested, waiting in Drive folder "nl-formation-runs (repo draft, 4 Oct 2026)"; CBH creates an empty public repository under NavigatorsLog and picks its license (GPL-3.0 suggested); decide whether to raise the units issue (confirmed against [102]'s own definition) with the collision module's author; hosting is decided [CBH]: GitHub Actions for runs, Colab for analysis; then set up the public repository, run exploratory benchmark B0, freeze and certify P-D12, and run it; in parallel, find a verified mechanism source for scenario B's iron gradient; then block 5 (thick-CO2 coefficients, single-shell melt fraction).

Decision markers

[CBH] marks a decision made by Christopher Blake Head. [CL] marks a decision or proposal that originated with Claude and stands until CBH confirms, amends, or strikes it. Superseded claims are struck through, not deleted, and logged in the revision log. Confirmation [CBH, 3 October 2026]: every [CL] item up to Checkpoint 5 is confirmed by CBH. The [CL] markers stay in place so the origin of each decision remains visible; new [CL] items after Checkpoint 5 still need review.

Purpose

Build an interactive model of a single rocky planet whose water moves between reservoirs from formation to the present, driven by thermodynamics, interior evolution, and the influence of the giant planets. The planet has tunable inputs. The model is credible only if, with its rules frozen

and its inputs taken from independent measurements, it reproduces the observed outcomes of the solar system's rocky planets. Where it fails, the failure is reported.

Project goal

Build an interactive, tunable model of a rocky planet's water from formation to the present, driven by thermodynamics, interior evolution, and giant-planet influence. With frozen rules and only measured inputs, it should reproduce the observed water histories of the solar system's rocky planets, label every debated or unverified input, and report its own misses. The visuals are surreal, but every visual property maps to real data. [Drafted CL, approved CBH 3 October 2026]

Deep-time caveat

This model spans about 4.5 billion years. Over that span it is our best guess, built from present-day measurements and a sparse record of the past. Passing the primary tests would not show that the model's path through the intervening billions of years is right; it would show only that the path ends in the right places and passes through the few dated windows we can check. [CBH]

Test coverage over time [CL]: 8 of the 14 primary tests check present-day state only (V1, V2, V4, E2, E4, M1, M2, M4). 4 check present-day quantities that record the whole history, such as D/H ratios and total loss to space (V3, E3, M3, M5). 2 check specific past windows (E1, M6). Mercury tests Me1 and Me2, once approved, add 2 more present-day checks. Formation tests F1 to F5, once approved, check present-day measurements that record the formation era.

How the model shows this [CL proposals]:

- Outputs are drawn as ranges, not single lines, and the ranges widen with distance into the past.
- Each era is labeled with the evidence that constrains it, so stretches with no direct data are visible as such.
- In the visuals, older eras render softer and more diffuse, matching the smooth versus sharp encoding if confidence is chosen for it.

Process principle [CBH]

Change through time is driven by the actual physical and chemical processes, each modeled on its own, never by fitting a curve to the outcome. The model records how systems affect each other, both directly and through intermediate systems. Processes that look the same at one scale often separate under closer or wider observation, and those differences decide how each reservoir holds onto its water or loses it. The model keeps such processes as separate fluxes rather than merging them. [CBH]

Applied to the current design [CL proposals]:

- Escape to space stays split by mechanism: thermal (Jeans) escape [65], diffusion-limited supply [66], energy-limited early escape [67], sputtering $\diamond 53$, the proposed HCO⁺ channel at Venus $\diamond 56$, and photochemical escape and ion pickup (sources still to be added). Each scales differently with the Sun, gravity and the atmosphere, so merging them would hide why Venus, Earth and Mars diverge.
- Water leaving the surface stays split by destination: lost to space versus locked in crustal

minerals ±29. Both shrink the surface inventory, but only escape raises D/H, which is what lets the D/H tests tell them apart.

- Carbonate formation stays split by pathway: surface weathering in the carbonate-silicate feedback [20] versus groundwater reacting with olivine underground, as at Jezero 44. Only the first acts as a climate thermostat.
- Couplings are written as an explicit table of which process feeds which, with a sign (amplifies or damps) and a reference, so indirect effects can be traced step by step.
- Each planet's water outcome is reported with the mechanisms that kept or removed its water, so the differences between Venus, Earth and Mars are explained by process, not only by end state.

Formation principle [CBH]

Planets affect each other, so their formation affected each other. Formation is a tested output for all four rocky planets, not a fixed input, and the four are formed from one shared, coupled history rather than one at a time. [CBH, 3 October 2026]

Consequences [CL proposals, ~~pending~~ CBH approved by CBH 3 October 2026]:

- Stage A becomes a coupled formation stage: one shared supply of planet-building material, shaped by the giant-planet scenario, feeds all four planets. Material taken up by one planet is not available to another.
- One published view of this coupling: if Mars and Mercury are embryos that avoided being absorbed, out of about 20 original embryos with the rest accreted into Venus and Earth, then mantle stripping of one of them by hit-and-run collisions is statistically probable ±95.
- Planet mass moves from tunable input to tested output, which changes the input list under D5. Orbital distance stays an input in v0.1.
- New formation tests, to be drafted with verified values: each planet's mass [91]; core size where measured (Mercury liquid core top at 2020 ± 30 km [93]; Mars ±37, ±46, ±47, ±48; Venus ±74); and volatile retention (Mercury's potassium is inconsistent with formation by extreme heating [45]).
- Mercury formation scenarios shown as debated: single giant impact, hit-and-run, multiple collisions, and accretion from chondrite-like material ±94, ±95, ±96, [45].
- Limit: ref 94 states that existing formation models lack the detail to be tested against the observations. The box version here is cruder, so formation tests may be weak and will be labeled that way.

New references, checked 3 October 2026: 93 Hauck, S.A. et al. (2013) The curious case of Mercury's internal structure. JGR Planets 118(6):1204-1220 (V*, DOI not retrieved). 94 Ebel, D.S., Stewart, S.T. The elusive origin of Mercury, in Mercury: The View after MESSENGER, arXiv:1712.08234 (V*, year and pages not confirmed). 95 doi:10.1038/ngeo2189, Nature Geoscience, hit-and-run origin of Mercury; attributed to Asphaug and Reufer (2014) by citing reviews (V*, title and authors not read from the paper itself). 96 Chau, A., Reinhardt, C., Helled, R., Stadel, J. (2018) Forming Mercury by giant impacts, arXiv:1808.02448 (V*, journal not confirmed).

Decisions log

#	Decision	By	Basis
D1	Build a tunable planet	[CBH]	More accurate; lets the

#	Decision	By	Basis
	rather than a single-planet timeline.		same rules be tested on several planets.
D2	Do not limit the test set to Venus, Earth, and Mars. Include giant-planet formation as an upstream driver of the rocky planets unless the concentration of evidence says otherwise.	[CBH]	Evidence check in section "Giant planet evidence check" supports inclusion.
D3	Mark debated studies with a symbol and reference number wherever they are used. Keep the symbol key and reference numbers in the evidence catalog at minimum.	[CBH]	Honest identification of knowledge gaps.
D4	Use data published by NASA and other space professionals, drawing on diverse, well-ranged measurements from across the solar system.	[CBH]	
D5	Freeze and hash the model rules before any test run. Inputs come only from independent measurements; outputs are never tuned toward the observed answer. Misses are published.	[CL]	Prevents curve-fitting with six or more knobs and few test planets.
D6	Use a box model (reservoirs connected by fluxes) built on published ranges, not a 3D climate simulation.	[CL]	Fits free-tier compute and the zero-cash mandate; a 3D model cannot be validated at this scale anyway.
D7	Represent giant-planet history as a scenario selector drawn from published dynamical simulations rather than	[CL]	The competing scenarios are themselves debated (see refs 2, 3, 5); a selector shows how

#	Decision	By	Basis
	running our own N-body simulations.		outcomes depend on that choice.
D8	Exclude icy moons (Titan, Europa, Enceladus) from the test set in v0.1.	[CL]	Different chemistry and heat sources from rocky-planet water. Open to reversal.
D9	Exclude exoplanets from v0.1.	[CL]	Observables are too sparse to test water history. Candidate for a later version.

D10 [CBH, 3 October 2026]: the model computes rocky-planet accretion with its own small N-body simulations from mechanism inputs only, so formation tests F1 and F2 stay independent (row not added to the table above; recorded here). Feasibility test and open design choices are in the Stage 2 document.

D11 [CBH, 4 October 2026]: building-block iron runs as two named scenarios, the same iron fraction everywhere (A) and iron varying with distance from a published mechanism (B), with test F3 reported for each. Scenario B needs a verified mechanism source that gives a number before it can run. Details are in the Stage 2 document.

Giant planet evidence check

Result: the evidence supports including the giant planets [CL assessment, for CBH review]. Meteorites split into two isotopically distinct families, an inner (non-carbonaceous) and an outer (carbonaceous) group, which is well established [1, 13]. Early formation of Jupiter is the leading explanation for keeping these reservoirs apart ± 1 . Mars grew to near its final size within a few million years and stopped [14], consistent with its growth being cut short. Earth received outer-solar-system material late in its accretion [13], which bears directly on where its water came from.

What is debated is the mechanism, not whether the giant planets mattered: an inward-then-outward migration of Jupiter ± 2 , an early giant-planet instability ± 5 , and inward scattering of water-rich bodies during giant-planet growth ± 6 are competing or partly overlapping explanations, and the timing of the instability is itself contested ± 3 .

Model structure

Stage A: disk and giant planets (upstream)

Sets the planet's starting water and its delivery history. Inputs: giant-planet scenario selector, planet's formation distance. Outputs: initial water inventory and its hydrogen isotope signature, timing of late delivery.

Stage B: interior

Magma ocean, outgassing of water and CO₂, core dynamo lifetime, tectonic style (moving plates versus a single rigid shell).

Stage C: climate and water cycle

Solar luminosity over time, greenhouse warming, the carbonate-silicate feedback that regulates CO₂ on Earth, and exchange among surface water, ice, vapor, and groundwater, including hydrothermal circulation.

Stage D: loss and storage

Escape of hydrogen and oxygen to space, and water locked into crustal minerals.

Reservoirs tracked

Water: surface liquid, ice, atmospheric vapor, groundwater and hydrothermal fluid, hydrated crustal minerals, mantle, lost to space. Carbon: atmospheric CO₂, carbonate rock, mantle. Tracked isotope: hydrogen (D/H), because it records water lost to space and can be checked against measurements at Venus, Earth, and Mars.

Tunable inputs

Input	Why it matters	Main evidence
Planet mass (moved to tested output by the Formation principle)	Gravity, interior cooling rate, dynamo and tectonic lifetime	36, 37
Distance from the Sun	Energy received, position relative to the snow line	17
Giant-planet scenario	Starting water and delivery history	‡2, ‡3, 4, ‡5, ‡6
Starting water inventory	Total water available	7, ‡10, ◇11, ◇12, ‡15
Dynamo lifetime	Possible shielding of the atmosphere	31, ‡32, ‡33, ‡34, ‡35
Tectonic style	Carbon recycling and climate stability	20, 36
Young Sun's ultraviolet output	Drives upper-atmosphere escape	18, 19

Test set

Body	Role	What is compared
Venus	Outcome test	D/H [21], present atmosphere, absence of surface water, escape rates [22]
Earth	Outcome test and reference	Ocean mass, D/H reference, early liquid water ‡16
Mars	Outcome test	D/H [25, 26], present loss rates [27, 28], crustal water ‡29, surface water record [42, ◇44]
Jupiter and Saturn	Upstream drivers, not outcome	Formation timing [1], migration

Body	Role	What is compared
	tests	and instability scenarios
Uranus and Neptune	Upstream drivers	Role in the giant-planet instability ± 3
Asteroid belt and meteorites	Starting materials	Water content and D/H of meteorite families [7, 13]
Comets	Delivery source check	D/H range [8, 9]
Moon	Impact history constraint	Dating of the bombardment record ± 3
Mercury	Candidate outcome test, pending CBH decision Outcome test [CBH, 3 October 2026]	Surface volatile content [45]; polar water ice [92]; exosphere pressure [91]

Validation protocol

Rules, equations, and parameter ranges are written into this spec and frozen with a hash before the first run [CL]. Each test body is run with its measured physical inputs only. Pass or fail tolerances for each observable must be set before the run. ~~Tolerances: proposed below, pending CBH.~~ Tolerances: approved by CBH 3 October 2026; see below. A miss is recorded with its likely cause and kept in the record.

Pass or fail tolerances [CL proposal, ~~pending CBH~~ approved CBH 3 October 2026]

~~These tolerances must be confirmed or amended by CBH and then frozen with the rules before the first run.~~ These tolerances were approved by CBH on 3 October 2026 and are frozen with the rules before the first run. The band widths are Claude's judgement, not values taken from the literature: no source in the catalog states how closely a box model of this kind should match each observable. [CL]

Rules

- Primary tests use only observables with no debated ($\pm$) status. Debated observables are reported as consistent or inconsistent but do not decide pass or fail. [CL]
- Continuous quantities are tested in factors (a band above and below the measured value), because a box model is expected to match orders of magnitude, not decimals. Bands are wider where the measured value itself swings with season. [CL]
- A planet passes when at least one input set inside the published input ranges passes all of its primary tests. The share of the input range that passes is also reported, so a narrow lucky fit is visible. [CL]
- ~~The solar-system test passes only if Venus, Earth and Mars~~ Venus, Earth, Mars and Mercury (Mercury added by CBH decision, 3 October 2026) all pass with one shared rule set. Inputs and rules may not be changed after results are seen; any change starts a new frozen version. [CL]
- Every miss is recorded with its likely cause and kept in the record (D5).

Primary tests

Measured values come from the catalog. Ref 91 is the NASA NSSDCA Planetary Fact Sheet (NASA Goddard Space Flight Center), checked 3 October 2026.

Test	Planet	Measured value	Pass band [CL]	Ref
V1	Venus	No surface liquid water today	Model shows no surface liquid water	22
V2	Venus	Surface pressure 92 bar	46 to 184 bar (factor 2)	91
V3	Venus	Water D/H about 100 times Earth's	50 to 200 times (factor 2)	21
V4	Venus	Present escape in the 0.5 to 2 kg/s range shared by Venus, Earth and Mars	0.2 to 5 kg/s	34
E1	Earth	Oceans for nearly 4 billion years	Surface liquid water present from 4.0 Ga to today	24
E2	Earth	Surface pressure 1 bar	0.5 to 2 bar (factor 2)	91
E3	Earth	Ocean water D/H (1.558 ± 0.001) $\times 10^{-4}$	1.04 to 2.34×10^{-4} (factor 1.5)	9
E4	Earth	Present escape in the shared 0.5 to 2 kg/s range	0.2 to 5 kg/s	34
M1	Mars	Cold and dry today	No stable surface liquid water today	27
M2	Mars	Surface pressure about 0.01 bar	0.003 to 0.03 bar (factor 3, seasonal swing)	91

M3	Mars	Water D/H 5 to 10 times Earth's ocean value	5 to 10 times	29
M4	Mars	Present hydrogen and oxygen escape about 2 to 3 kg/s	0.7 to 9 kg/s (factor 3, seasonal swing)	27, 28
M5	Mars	Integrated water loss to space up to 23 m global layer, stated as likely a lower limit	At least 23 m	27
M6	Mars	Lakes and rivers 3.5 to 3.8 billion years ago	Surface liquid water present at least episodically between 3.8 and 3.5 Ga	24

Mercury tests [CL proposal, pending CBH approved by CBH 3 October 2026]

- Me1, Mercury: surface pressure below about 5×10^{-15} bar, essentially a vacuum [91]. Pass band [CL]: the model retains no collisional atmosphere today (surface pressure below 10^{-9} bar). A box model cannot represent a surface-bounded exosphere, so this test checks only that no real atmosphere survives. Confidence [CL]: measurement high (NASA upper limit); diagnostic power low, because nearly any small, hot model planet passes. Kept as a sanity check.
- Me2, Mercury: water ice in permanently shadowed polar craters today [92]. Pass band [CL]: the model holds a nonzero polar cold-trap ice reservoir today. Confidence [CL]: measurement high (radar, neutron and laser-reflectance evidence agree); diagnostic power medium. Caveat: a global box model has no poles, so this needs a polar cold-trap sub-reservoir. Only presence is tested because the ice's origin is debated.

Ref 92: Lawrence, D.J. et al. (2013) Evidence for water ice near Mercury's north pole from MESSENGER Neutron Spectrometer measurements. Science 339(6117). ~~V* (DOI not retrieved)~~. Science 339:296, doi:10.1126/science.1229764 (V; found during Stage 2 block 2).

Ref 91 now also covers NASA's Mercury Fact Sheet. No Mercury water D/H measurement was found in the catalog sources, so no isotope test is proposed.

Formation tests [CL proposal, pending CBH approved by CBH 3 October 2026]

- F1, all four planets: masses 0.330 (Mercury), 4.87 (Venus), 5.97 (Earth) and 0.642 (Mars) $\times 10^{24}$ kg [91]. Pass band [CL]: each within a factor of 2. Confidence [CL]:

measurement very high; diagnostic power medium, because a factor of 2 is loose. Valid only under the independence safeguard below.

- F2, Mars and Mercury small: Mars is about 0.11 and Mercury about 0.055 of Earth's mass [91]. Pass band [CL]: both below 0.15 Earth masses while Venus and Earth form near their measured masses. This is the small-Mars outcome that giant-planet scenarios are built to explain. Confidence [CL]: measurement very high; diagnostic power high, the strongest formation test. Valid only under the independence safeguard below.
- F3, Mercury core: top of the liquid core at 2020 ± 30 km of a 2440 km radius [93]. Pass band [CL]: 1820 to 2220 km (within 10 percent). Confidence [CL]: measurement high; the value is inferred from moment-of-inertia data through interior models, and an independent MESSENGER team estimate of 2030 ± 37 km (NASA technical record 20120009897) falls well inside the band. Diagnostic power high, because Mercury's oversized core is the feature formation scenarios must explain.
- F4, Earth core: core-mantle boundary at 3480 km radius [97]. Pass band [CL]: 3130 to 3830 km (within 10 percent). Confidence [CL]: measurement very high (global seismology); diagnostic power medium, because Earth is the reference case.
- F5, Mercury volatile retention: surface potassium relative to thorium is inconsistent with formation by extreme heating [45]; Mercury's K/Th sits with the other terrestrial planets near 5500, not with the Moon near 360 [98]. Pass band [CL]: the model's Mercury keeps moderately volatile elements at the terrestrial-planet level, not depleted to the lunar level. This needs a potassium tracer carried through formation. Confidence [CL]: measurement medium (gamma-ray data from Mercury's northern hemisphere only, potassium uncertainty about 20 percent); interpretation debated, since some studies argue giant impacts need not remove volatiles ± 94 . Diagnostic power medium. Requires a potassium tracer.

Ref 97: Dziewonski, A.M., Anderson, D.L. (1981) Preliminary reference Earth model. *Physics of the Earth and Planetary Interiors* 25(4):297-356. doi:10.1016/0031-9201(81)90046-7 (V).

Ref 98: MESSENGER Science Highlight, The MESSENGER Gamma-Ray Spectrometer: a window into the formation and early evolution of Mercury, Johns Hopkins APL, 20 December 2011 (V). Ref 91 now also covers planetary masses from the NASA fact sheet.

Confidence ratings [CL]: measurement confidence is how well the observed value is known.

Diagnostic power is how strongly a pass or a fail would separate competing formation or evolution scenarios. Ratings are judgement calls based on the cited sources.

Independence safeguard [CL]: the giant-planet and formation scenario selectors import mechanism parameters from published simulations (for example where the disk was depleted or truncated), never the simulations' final planet masses or core sizes. Otherwise F1 to F3 would test the model against its own inputs.

Secondary observables (reported, not pass or fail)

- Mars starting water of 100 to 1500 m global equivalent layer and 30 to 99 percent crustal sequestration ± 29 .
- Mars northern ocean ± 26 , ± 41 , $\diamond 64$.
- Mars mid-crust liquid water ± 30 , ± 63 .
- Whether Venus ever had oceans ± 23 , ± 24 , $\diamond 55$.
- Mars core radius ± 46 , ± 47 , ± 48 , and when the Mars dynamo stopped ± 32 , ± 33 , $\diamond 49$.
- Order of the Jezero water episodes (groundwater, lake, hydrothermal) $\diamond 44$: too local for a global box model, so recorded as a qualitative check.

- Mercury polar ice mass of 2.1×10^{13} to 1.4×10^{15} kg: seen only in a secondary source quoting ref 92, so not used for pass or fail. Whether the ice was delivered or outgassed is debated; sources still to be added.
- Mars core size ‡37, ‡46, ‡47, ‡48 and Venus core state ‡74: debated or unmeasured, so reported as formation context rather than tested.

Data sources

International public data [CBH]: draw on public satellite and mission data from agencies worldwide, not only NASA. Candidates include ESA, JAXA, ISRO, CNSA, the UAE Space Agency, and Roscosmos archives, alongside NASA's Planetary Data System. Physics and propulsion research is also in scope [CBH]. A research pass on these sources is under way; findings enter the evidence catalog with reference numbers and symbols.

Reading of propulsion data [CL, ~~pending CBH confirmation~~ confirmed by CBH 3 October 2026]: interpreted as spacecraft tracking and trajectory data (radio science that yields planetary gravity fields, interior structure, and core size), plasma physics shared by electric propulsion and atmospheric ion escape, and escape-velocity physics. CBH adds that giant-planet migration is in scope and not moot; it is carried by the giant-planet scenario selector (D7) and refs ‡2, ‡3, ‡5, ‡50.

Tools and how they are used

- Web search and fetch: mission archives, journal pages, and DOIs. Main route for verifying references marked P. [CL]
- Consensus (peer-reviewed paper search): finds replications that move a $\diamond$ study to accepted or to ‡. [CL]
- NASA ADS (astronomy literature database): no connector exists; reached through web search. [CL]
- Google Docs and Drive: this spec, the decisions log, and the evidence catalog, edited in place. [CBH]
- Google Sheets: the parameter table (each input with its range, units, and reference number). The model reads from it so the spec and the code cannot drift apart. [CL]
- Code execution: a Python reference version of the box model, used to test the physics before visuals exist and to compute the freeze hash. [CL]
- Navigator's Log commit-before-run skill: preregistration with freeze-and-hash before any test run. [CL]
- Design canvas and Design System: visual mockups for iteration, and a locked visual language (palettes, smooth and sharp rules, type). [CL]
- Accessibility review skill: checks the visuals against seizure-risk and motion standards. [CL]
- Netlify: hosting. Candidate existing sites for reuse, judged by name only and not yet inspected: navigatorslog (host), geometricencoder, grisaille. [CL]
- Not used for this project: PubMed, bioRxiv, BioSkepsis (biomedical); Gmail, Calendar, Indeed. [CL]

Visual design

Direction [CBH]: surreal and trippy, inspired by real lava lamps, with variation between smooth and sharp visuals for depth, while remaining true representations of the data. Reuse of other deployed Netlify projects is open.

Basis [CL]: a lava lamp runs on buoyancy-driven convection, the same physics that moves a planet's mantle and atmosphere. Blob motion is therefore driven by the model's convection strength rather than added as decoration, and it slows as the planet cools.

Encoding rules [CL proposals]:

- Size equals amount: blob area is proportional to the water or carbon in that reservoir.
- Color equals temperature, on a perceptually even scale. No rainbow scales, which distort data.
- Flow speed equals flux between reservoirs.
- ~~Smooth versus sharp equals confidence: measured values render crisp and faceted; debated ($\pm$) values render soft and diffuse. Alternative under consideration: sharpness by physical state (liquids smooth, rock and ice faceted).~~
- Smooth versus sharp [CBH, 3 October 2026]: shape by physical state (liquids and gases smooth, rock and ice faceted); edge blur by confidence (measured values crisp, debated ($\pm$) values blurred, older eras softer per the Deep-time caveat).
- Truth layer: a toggle that shows the raw numbers and reference numbers beneath any visual.
- Safety: no flashing faster than three times per second (the seizure-risk threshold in accessibility standards), and a reduced-motion mode.

Rendering [CL]: WebGL shaders, checked against the Python reference model so the visuals cannot drift from the math.

Known knowledge gaps

- How warm and wet early Mars was: sustained warm climate versus mostly frozen with episodic melting [38], ± 39 , ± 40 .
- Whether Mars had a northern ocean, and its size ± 26 , ± 41 .
- How Mars's lost water splits between escape to space and crustal storage ± 29 , ± 30 .
- When Mars's dynamo stopped, with published estimates ranging from about 4.1 billion years ago ± 32 to activity at 3.7 billion years ago ± 33 .
- Whether a magnetic field protects an atmosphere at all ± 34 .
- Where Mars's carbon went (the missing carbonates) [42], ± 43 , $\diamond 44$.
- Whether Venus ever had oceans ± 23 , ± 24 .
- The young Sun's ultraviolet output, which depends on its unknown early rotation rate [19].
- Whether Earth's water was mostly local ± 10 , $\diamond 11$, ± 15 or delivered from the outer solar system [7, 13].
- Timing of the water-rock episodes at Jezero, which the authors state cannot yet be dated $\diamond 44$.
- Mars core radius and whether a molten silicate layer sits above the core: 1830 ± 40 km [37], 1650 ± 20 km with a molten layer ± 46 , about 1790 to 1850 km from radio tracking ± 48 . Ref 46 also states that the layered mantle requires external sources for the crustal magnetization, which bears on the dynamo input.
- Whether Venus's core is liquid ± 74 .

- Timing of the giant-planet instability: 60 to 100 Myr after the start of the Solar System ‡50, disputed by Izidoro and colleagues.
- Whether Earth had a magnetic field before 3.5 billion years ago ‡35, ‡84.

Evidence catalog

Symbol key

‡ Debated: competing published interpretations exist, or the result is contested in the literature.

◇ Single recent study: not yet independently replicated.

No symbol: broadly accepted, though values may carry wide ranges.

Numbers in brackets or after a symbol are reference numbers in the table below.

Verification status

V: checked against the source or its official release during this drafting session. **P**: cited from Claude's background knowledge; title, journal, and values must be checked against the source before the rules are frozen [CL]. As of 3 October 2026 all 45 original references are V. The addendum below uses V and V*.

Ref	Sym	Citation	Used for	Ver.
1	‡	Kruijer et al. (2017), Age of Jupiter inferred from the distinct genetics and formation times of meteorites, PNAS	Two meteorite families (accepted); Jupiter as the separator (debated)	V
2	‡	Walsh et al. (2011), Grand Tack model, Nature	Jupiter migration scenario; small Mars	V
3	‡	Tsiganis et al. (2005) and Gomes et al. (2005), Nice model, Nature	Giant-planet instability; timing contested	V
4		Nesvorny (2018), Dynamical evolution of the early Solar System, Annual Review of Astronomy and Astrophysics	Review of instability scenarios	V
5	‡	Clement et al. (2018), early giant-planet instability and	Alternative small-Mars scenario	V

Ref	Sym	Citation	Used for	Ver.
		Mars's growth, Icarus		
6	‡	Raymond and Izidoro (2017), Origin of water in the inner Solar System, Icarus	Water delivery by giant-planet growth	V
7		Alexander et al. (2012), provenances of asteroids and volatile inventories of the terrestrial planets, Science	Chondrite D/H matches Earth's water	V
8		Altwegg et al. (2015), D/H of comet 67P, Science	Comet D/H about three times Earth's	V
9		Hartogh et al. (2011), Ocean-like water in comet 103P/Hartley 2, Nature	Comet D/H varies	V
10	‡	Piani et al. (2020), hydrogen in enstatite chondrites, Science	Earth's water possibly local	V
11	◊	Grewal et al. (2024), Accretion of the earliest inner Solar System planetesimals beyond the water snowline, Nature Astronomy	Earliest inner bodies held water	V
12	◊	Grewal, Zhang and Drazkowska (2026, published 18 September), Nature Astronomy	Earliest outer bodies were 83 to 92 percent chondrules, with little icy dust	V
13		Budde, Burkhardt and Kleine (2019), molybdenum isotopic evidence for late accretion	Late outer-system delivery to Earth	V

Ref	Sym	Citation	Used for	Ver.
		of outer Solar System material to Earth, Nature Astronomy		
14		Dauphas and Pourmand (2011), Mars formed rapidly, Nature	Mars as a stranded embryo	V
15	‡	Young et al. (2023), Earth shaped by primordial H ₂ atmospheres, Nature	Water made from nebular hydrogen and magma ocean	V
16	‡	Wilde et al. (2001), 4.4-billion-year-old zircons, Nature	Early liquid water on Earth	V
17		Gough (1981), Solar interior structure and luminosity variations, Solar Physics	Faint young Sun	V
18		Ribas et al. (2005), Sun in Time, Astrophysical Journal	Ultraviolet output history	V
19		Tu et al. (2015), extreme ultraviolet and X-ray Sun in time, Astronomy and Astrophysics	Ultraviolet history depends on early rotation	V
20		Walker, Hays and Kasting (1981), Journal of Geophysical Research	Carbonate-silicate feedback	V
21		Donahue et al. (1982), Venus was wet, Science	Venus D/H about 100 times Earth's	V
22		Barabash et al. (2007), ion loss from Venus (Venus Express), Nature	Present Venus escape	V
23	‡	Way et al. (2016), habitable early	Venus oceans possible	V

Ref	Sym	Citation	Used for	Ver.
		Venus, Geophysical Research Letters		
24	‡	Turbet et al. (2021), Venus never condensed oceans, Nature	Venus oceans unlikely	V
25		Webster et al. (2013), Curiosity isotope ratios, Science	Mars D/H about 6 times Earth's Value not confirmed in retrieved source; use 5 to 10 × SMOW [29].	V
26	‡	Villanueva et al. (2015), Mars water isotopes, Science	D/H map; minimum early ocean volume (debated)	V
27		Jakosky et al. (2018), MAVEN loss rates and integrated loss, Icarus	Mars escape now and through time	V
28		Chaffin et al. (2017), high-altitude water and hydrogen escape, Nature Geoscience	Seasonal escape	V
29	‡	Scheller et al. (2021), sequestration of water in Mars's crust, Science	30 to 99 percent of water stored in crust	V
30	‡◊	Wright, Morzfeld and Manga (2024), liquid water in Mars's mid-crust, PNAS	Deep groundwater from InSight seismic data	V
31		Acuna et al. (1999), Mars crustal magnetization, Science	Early Mars dynamo existed	V
32	‡	Vervelidou et al. (2017),	Shutoff about 4.12 to 4.14 billion	V

Ref	Sym	Citation	Used for	Ver.
		Constraining the date of the Martian dynamo shutdown by means of crater magnetization signatures, JGR Planets	years ago	
33	‡	Mittelholz et al. (2020), timing of the Martian dynamo, Science Advances	Dynamo active 4.5 and 3.7 billion years ago	V
34	‡	Gunell et al. (2018), why an intrinsic magnetic field does not protect a planet against atmospheric escape, Astronomy and Astrophysics	Magnetic shielding contested	V
35	‡	Tarduno et al. (2015), Earth's magnetic field 4.2 billion years ago, Science	Early Earth dynamo	V
36		Grott et al. (2013), long-term evolution of the Martian crust-mantle system, Space Science Reviews	Mars single rigid shell (stagnant lid)	V
37	‡	Stahler et al. (2021), InSight seismic detection of Mars's core, Science	Mars interior structure Core radius 1830 ± 40 km is contested: see ‡46, ‡47, ‡48.	V
38		Wordsworth (2016), The climate of early Mars, Annual Review of Earth and Planetary Sciences	Warm versus cold early Mars review	V

Ref	Sym	Citation	Used for	Ver.
39	‡	Ramirez et al. (2014), warming early Mars with CO ₂ and H ₂ , Nature Geoscience	Hydrogen-assisted greenhouse	V
40	‡	Wordsworth et al. (2017), transient reducing greenhouse on early Mars, Geophysical Research Letters	Episodic warming	V
41	‡	Di Achille and Hynek (2010), ancient ocean on Mars, Nature Geoscience	Northern ocean	V
42		Ehlmann et al. (2008), carbonates in Nili Fossae, Science	Carbonate near Jezero	V
43	‡	Hu et al. (2015), tracing the fate of carbon on Mars, Nature Communications	Carbon loss to space versus rock	V
44	◇	Bedford et al. (2026), Lake- and groundwater-associated alteration of the olivine-rich Margin unit in Jezero crater, Mars, Communications Earth and Environment 7, doi:10.1038/s43247-026-03997-9	Three ordered water episodes (groundwater, lake, hydrothermal); carbonate formation; dates unknown	V
45		Peplowski et al. (2011), MESSENGER potassium on Mercury, Science	Mercury volatile content	V

Evidence catalog addendum

References 46 to 90 were added on 3 October 2026 after a verification pass against retrieved sources. Ver. column: V means author, title, journal, year and identifier were confirmed; V* means the citation was confirmed but the DOI, volume, pages or full title could not be retrieved. Symbols follow the key above. [CL]

Ref	Sym	Citation	Used for	Ver.
46	‡	Samuel, H. et al. (2023) Geophysical evidence for an enriched molten silicate layer above Mars's core. Nature 622:712-717. doi:10.1038/s41586-023-06601-8	Mars core radius 1650 ± 20 km beneath a molten silicate layer; layered mantle needs external sources for crustal magnetization	V
47	‡	Khan, A. et al. (2023) Evidence for a liquid silicate layer atop the Martian core. Nature 622:718-723. doi:10.1038/s41586-023-06586-4	Liquid silicate layer about 150 km thick above the core	V
48	‡	Le Maistre, S. et al. (2023) Spin state and deep interior structure of Mars from InSight radio tracking. Nature 619:733-737. doi:10.1038/s41586-023-06150-0	No solid inner core; core radius about 1790 to 1850 km from radio tracking	V
49	◇	Steele, S.C. et al. (2024) Weak magnetism of	Mars dynamo possibly active and reversing	V

		Martian impact basins may reflect cooling in a reversing dynamo. Nature Communications 15:6831. doi:10.1038/s41467-024-51092-4	until 3.9 to 3.7 Ga	
50	‡	Avdellidou, C., Delbo, M., Nesvorny, D., Walsh, K.J., Morbidelli, A. (2024) Dating the Solar System's giant planet orbital instability using enstatite meteorites. Science 384:348-352. doi:10.1126/science.adg8092	Giant-planet instability 60 to 100 Myr after Solar System formation; disputed by Izidoro and colleagues	V
51	◇	Kleinböhl, A. et al. (2024) Hydrogen escape on Mars dominated by water vapour photolysis above the hygropause. Nature Astronomy 8:827-837. doi:10.1038/s41550-024-02268-x	Source of escaping hydrogen at Mars	V
52		Montmessin, F. et al. (2024) Mars' water cycle and	Review of Mars water cycle and escape data	V

		escape: a view from Mars Express and beyond. Space Science Reviews 220(7). doi:10.1007/s11214-024-01099-6		
53	◇	Curry, S.M. et al. (2025) First direct observations of atmospheric sputtering at Mars. Science Advances 11(22). doi:10.1126/sciadv.adt1538	Sputtering escape observed directly	V
54	‡	Clarke, J.T. et al. (2024) Martian atmospheric hydrogen and deuterium: seasonal changes and paradigm for escape to space. Science Advances 10:eadm7499. doi:10.1126/sciadv.adm7499	Seasonal H and D escape; D/H fractionation varies with season	V
55	◇	Constantinou, T., Shorttle, O., Rimmer, P.B. (2024) A dry Venusian interior constrained by atmospheric chemistry. Nature Astronomy.	Dry Venus interior; argues against past oceans	V

		doi:10.1038/s41550-024-02414-5		
56	◇	Chaffin, M.S. et al. (2024) Venus water loss is dominated by HCO ⁺ dissociative recombination. Nature 629:307-310. doi:10.1038/s41586-024-07261-y	Proposed dominant Venus hydrogen loss channel; HCO ⁺ not yet measured at Venus	V
57		Glavin, D.P., Dworkin, J.P. et al. (2025) Abundant ammonia and nitrogen-rich soluble organic matter in samples from asteroid (101955) Bennu. Nature Astronomy 9:199-210. doi:10.1038/s41550-024-02472-9	Bennu volatile-rich; parent body accreted outer Solar System ices	V
58		McCoy, T.J. et al. (2025) An evaporite sequence from ancient brine recorded in Bennu samples. Nature 637:1072-1077	Brine on Bennu's parent body	V*
59		Zega, T.J. et al. (2025)	Alteration near 25 °C, similar to	V*

		Mineralogical evidence for hydrothermal alteration of Bennu samples. Nature Geoscience 18:832-839	Ryugu and CI chondrites	
60	◇	Barrett, T.J., Bryson, J.F.J., Geraki, K. (2025) The source of hydrogen in earth's building blocks. Icarus 116588. doi:10.1016/j.icarus.2025.116588	Hydrogen bonded to sulfur in enstatite chondrite matrix	V
61	◇	Miozzi, F. et al. (2025) Experiments reveal extreme water generation during planet formation. Nature 648:551-555. doi:10.1038/s41586-025-09816-z	Water generated by hydrogen reacting with magma	V
62	◇	He, H., Li, L., Hu, S. et al. (2025) Water abundance in the lunar farside mantle. Nature 643:366-370. doi:10.1038/s41586-025-08870-x	Lunar farside mantle 1 to 1.5 µg/g water	V
63	‡	Challenges and reply on ref 30:	Mid-crust liquid water on Mars	V

		Jakosky et al. (2025) Results from the InSight Mars mission do not require a water-saturated mid crust, PNAS, doi:10.1073/pnas.2418978122; Xiao et al. (2025) Liquid water might not be the only answer, PNAS, doi:10.1073/pnas.2503071122; Wright et al. (2025) reply, PNAS, doi:10.1073/pnas.2505168122	not required by InSight data	
64	◇	Li, J., Liu, H. et al. (2025) Ancient ocean coastal deposits imaged on Mars. PNAS 122(9):e2422213122. doi:10.1073/pnas.2422213122	Zhurong radar evidence for an ancient coastline	V
65		Chamberlain, J.W. (1963) Planetary coronae and atmospheric evaporation. Planetary and Space Science 11:901-960	Jeans escape theory	V
66		Hunten, D.M. (1973) Journal of the	Diffusion-limited escape	V*

		Atmospheric Sciences 30:1481 (start page 1481 or 1482 unresolved)		
67		Watson, A.J., Donahue, T.M., Walker, J.C.G. (1981) Icarus 48:150-166	Energy-limited and rapid hydrodynamic escape	V*
68		Stevenson, D.J., Spohn, T., Schubert, G. (1983) Magnetism and thermal evolution of the terrestrial planets. Icarus 54:466-489	Dynamo criterion and core cooling	V
69		Solomatov, V.S., Moresi, L.-N. (2000) Scaling of time-dependent stagnant lid convection: application to small-scale convection on Earth and other terrestrial planets. JGR 105. doi:10.1029/2000JB900197	Stagnant-lid convection scaling	V
70		Reese, C.C., Solomatov, V.S., Moresi, L.N. (1998) Heat transport efficiency for	Stagnant-lid heat transport on Mars and Venus	V*

		stagnant lid convection with dislocation viscosity: application to Mars and Venus. JGR Planets		
71		Solomatov, V.S., Moresi, L.N. (1996) Stagnant lid convection on Venus. JGR Planets	Venus mantle viscosity 10^{20} to 10^{21} Pa s	V*
72		Keating, G.M. et al. (1998) The structure of the upper atmosphere of Mars: in situ accelerometer measurements from Mars Global Surveyor. Science 279:1672-1676. doi:10.1126/science.279.5357.1672	Mars thermosphere density from aerobraking	V
73		Müller-Wodarg, I.C.F., Bruinsma, S., Marty, J.-C., Svedhem, H. (2016) In situ observations of waves in Venus's polar lower thermosphere with Venus Express aerobraking. Nature Physics. doi:10.1038/nphys3733	Venus polar thermosphere density from aerobraking	V

74	‡	Konopliv, A.S., Yoder, C.F. (1996) Venusian k2 tidal Love number from Magellan and PVO tracking data. Geophysical Research Letters 23. doi:10.1029/96GL01589	Venus k2 = 0.295 ± 0.066 ; liquid-core reading contested	V
75		Genova, A. et al. (2019) Geodetic evidence that Mercury has a solid inner core. Geophysical Research Letters 46(7):3625-3633 . doi:10.1029/2018GL081135	Mercury solid inner core 0.3 to 0.7 of outer core radius	V
76		Zuber, M.T. et al. (2013) Gravity field of the Moon from the Gravity Recovery and Interior Laboratory (GRAIL) mission. Science 339:668. doi:10.1126/science.1231507	Lunar gravity field	V
77		Wieczorek, M.A. et al. (2013) The crust of the Moon as seen by GRAIL. Science 339:671-675.	Lunar crust density 2550 kg per cubic meter, porosity 12%	V

		doi:10.1126/science.1231530		
78		less, L. et al. (2018) Measurement of Jupiter's asymmetric gravity field. Nature 555:220-222. doi:10.1038/nature25776	Jupiter gravity field and wind depth (not the dilute core)	V
79		Wahl, S.M. et al. (2017) Comparing Jupiter interior structure models to Juno gravity measurements and the role of a dilute core. Geophysical Research Letters 44(10):4649	Jupiter dilute core; 7 to 25 Earth masses of heavy elements	V
80		less, L. et al. (2019) Measurement and implications of Saturn's gravity field and ring mass. Science 364:eaat2965. doi:10.1126/science.aat2965	Saturn core 15 to 18 Earth masses; winds at least 9000 km deep	V
81	◇	Mankovich, C., Fuller, J. (2021) A diffuse core in Saturn revealed by ring seismology. Nature	Saturn diffuse core to about 60% of radius, about 17 Earth masses of ice and rock	V*

		Astronomy		
82	‡	Lis, D.C. et al. (2019) Terrestrial deuterium-to-hydrogen ratio in water in hyperactive comets. Astronomy and Astrophysics 625:L5. doi:10.1051/0004-6361/201935554	Comet 46P D/H (1.61 ± 0.65) $\times 10^{-4}$; hyperactive comets may carry Earth-like water	V
83		Margot, J.-L. et al. (2021) Nature Astronomy 5:676-683. doi:10.1038/s41550-021-01339-7	Venus moment of inertia 0.337 ± 0.024	V*
84	‡	Challenge and defence of ref 35: Tarduno, J.A. et al. (2020) PNAS 117(5):2309 (defence); Direct age constraints on the magnetism of Jack Hills zircon, Science Advances, doi:10.1126/sciadv.add1511 (challenge)	Whether Earth had a field before 3.5 Ga	V*
85		Hood (2026) Regional mapping of MAVEN orbital magnetometer	Mars dynamo duration	V

		data: implications for the nature of crustal field sources and the duration of the Mars dynamo. JGR Planets. doi:10.1029/202 5JE009486		
86	◇	Srivastava et al. (2025) The late formation of chondrites as a consequence of Jupiter-induced gaps and rings. Science Advances 11(43):eady4823 . doi:10.1126/scia dv.ady4823	Jupiter shaping when chondrites formed	V
87	◇	Sirono et al. (2025) Chondrule formation by collisions of planetesimals containing volatiles triggered by Jupiter's formation. Scientific Reports 15:30919. doi:10.1038/s41 598-025-12643- x	Jupiter formation timing	V
88		Barnes, J.J., Nguyen, A.N. et al. (2025) The variety and	Materials accreted by Bennu's parent body	V

		origin of materials accreted by Bennu's parent asteroid. Nature Astronomy 9:1785-1802. doi:10.1038/s41550-025-02631-6		
89		Susarla et al. (2024) Variability of atomic hydrogen brightness in the Martian exosphere: insights from the Emirates Ultraviolet Spectrometer on board Emirates Mars Mission. JGR Space Physics. doi:10.1029/2024JA032525	Hope mission hydrogen corona	V
90		Chirakkil et al. (2024) EMM EMUS observations of hot oxygen corona at Mars: radial distribution and temporal variability. JGR Space Physics	Hope mission oxygen corona	V*

New references, checked 4 October 2026 [CL]: 99 Leinhardt, Z.M., Stewart, S.T. (2012) Collisions between gravity-dominated bodies. I. Outcome regimes and scaling laws. Astrophysical Journal 745:79, doi:10.1088/0004-637X/745/1/79, arXiv:1106.6084 (V). 100 Stewart, S.T., Leinhardt, Z.M. Collisions between gravity-dominated bodies, part 2, arXiv:1109.4588 (V*, full subtitle and journal not retrieved). 101 Chambers, J.E. (2013) Icarus 224:43, doi:10.1016/j.icarus.2013.02.015 (V*, title not read; journal, volume and DOI confirmed through two citing papers). 102 Childs, A.C., Steffen, J.H. (2022) Collisional fragmentation and

bulk composition tracking in REBOUND. MNRAS 511:1848, doi:10.1093/mnras/stac158, arXiv:2201.05714; code at github.com/annacrn/rebound_fragmentation (V). 103 Ferich, N., Childs, A.C., Steffen, J.H. (2024) Composition tracking for collisions between differentiated bodies in REBOUND, arXiv:2406.08588 (V*, journal not confirmed). 104 Tajer, H., Wang, J., Childs, A.C., Ferich, N., Lu, T., Rein, H. (2025) Origins of Mercury's big heart of iron: exploring pathways to form high core mass fraction (CMF) planets via N-body simulations, arXiv:2511.01842 (V*, preprint, journal not confirmed). 105 Lu, T., Tajer, H., Hernandez, D.M., Rein, H., Liu, A., Rice, M. (2025) Collisional fragmentation support in TRACE, arXiv:2505.04399 (V*, journal not confirmed). 106 Damer, B., Deamer, D. (2020) The hot spring hypothesis for an origin of life. Astrobiology 20(4):429-452, doi:10.1089/ast.2019.2045 (V). Context only, not a test input; see open items.

Excluded evidence (not used)

Rule [CBH]: evidence that cannot be confirmed against a retrievable source is not used. The following were checked and excluded:

- Chandrasekar et al., Planetary Science Journal (reversing Mars dynamo): no such paper found. The study is Steele et al. 2024 (ref 49); the name belongs to the author of a news article about it.
- A 2:1 ratio of escaping hydrogen to oxygen ions attributed to ref 22: not stated in the retrieved source. Ref 22 is kept for what it does state: the dominant escaping ions are O⁺, He⁺ and H⁺.
- A Mars D/H of about 6 times Earth's attributed to ref 25: not in the retrieved text. Use 5 to 10 × SMOW from ref 29.
- A revised Venus D/H of 150 ± 30 times Earth's (Donahue et al. 1997): seen only in a secondary source.
- Borlina et al. 2020 and Palin et al. 2024: citation details or stance not confirmed. The challenge to ref 35 is carried by ref 84; the challenge to ref 30 by ref 63.
- Lammer et al. 2003, Morschhauser et al. 2011, Yang et al. 2025: not confirmed and not needed. Energy-limited escape is covered by ref 67.
- Attribution correction: the research report credited the Jupiter dilute core to less et al. 2018, which measures Jupiter's asymmetric gravity field and wind depth (ref 78). The dilute core comes from Wahl et al. 2017 (ref 79).

Open items for CBH

- ~~Set pass or fail tolerances for each observable.~~
- ~~Review, then confirm or amend, the proposed pass or fail tolerances.~~
- ~~Decide whether Mercury joins the test set.~~
- ~~Confirm or strike each [CL] decision.~~
- ~~Choose the smooth versus sharp encoding: confidence or physical state.~~
- ~~Confirm or correct the reading of physics and propulsion data in Data sources.~~
- ~~Approve or amend Mercury tests Me1 and Me2.~~
- ~~Approve or amend the formation consequences [CL], including moving planet mass from tunable input to tested output.~~
- ~~Approve or amend formation tests F1 to F5.~~
- Draft, verify and approve the formation tests (mass, core size, volatile retention).

- Consider an origin-of-life context layer [idea raised by CBH; layer proposal CL, pending CBH]: the hot spring hypothesis $\pm 10^6$ places life's start in volcanic hot spring pools on land with repeated wet and dry cycles, as an alternative to deep-sea vents. The model already tracks volcanism, surface water and climate, but not how much land stands above sea level, so a view showing when exposed volcanic land and fresh surface water coexist (Earth, early Mars) would need a new land-area process. Context only, not a test.

Revision log

v0.1, 3 October 2026: spec started. No superseded claims yet.

v0.1, 3 October 2026, second entry: added Project goal, Data sources, Tools and how they are used, and Visual design sections. Corrected a statement made in conversation (not in this document) that the Google Docs editor connector was unavailable; it is available, so this document is edited in place.

v0.1, 3 October 2026, third entry: completed source verification. All 45 original references confirmed (V). Reference 37 marked debated; the $6\times$ value in reference 25 struck through. Added addendum references 46 to 90 and a list of excluded evidence. Corrected an attribution in the research report: the Jupiter dilute core comes from Wahl et al. 2017 (ref 79), not less et al. 2018 (ref 78).

v0.1, 3 October 2026, fourth entry: added proposed pass or fail tolerances [CL, pending CBH] with 14 primary tests across Venus, Earth and Mars, rules for debated observables, and ref 91 (NASA NSSDCA Planetary Fact Sheet). The open item on tolerances is struck through and replaced with a review item.

v0.1, 3 October 2026, fifth entry: CBH approved the pass or fail tolerances and rules; status wording updated, with the old wording struck through. Added the Deep-time caveat [CBH] after Project goal, with a test-coverage breakdown and display proposals [CL].

v0.1, 3 October 2026, sixth entry: added the Process principle [CBH] after the Deep-time caveat, with five design applications [CL].

v0.1, 3 October 2026, seventh entry: added Project status and stopping point section near the top, with a six-stage plan [CL]. This section is updated at every stopping point.

v0.1, 3 October 2026, eighth entry: CBH decided the smooth versus sharp encoding (both: shape by state, edge blur by confidence), made Mercury a full outcome test, and confirmed the propulsion-data reading while adding that giant-planet migration is in scope. The solar-system rule now includes Mercury. Proposed Mercury tests Me1 and Me2 [CL, pending CBH] and added ref 92. Superseded wording struck through; the status section was rewritten for Checkpoint 2.

v0.1, 3 October 2026, ninth entry: CBH decided that formation is a tested output for all four rocky planets, formed from one coupled history. Added the Formation principle section with consequences [CL, pending CBH] and refs 93 to 96. Status rewritten for Checkpoint 3.

v0.1, 3 October 2026, tenth entry: CBH approved the formation consequences, including planet mass moving from tunable input to tested output. Added proposed formation tests F1 to F5 [CL, pending CBH] and refs 97 and 98. Status rewritten for Checkpoint 4.

v0.1, 3 October 2026, eleventh entry: added confidence annotations [CL] to Me1, Me2 and F1 to F5, a rating key, and an independence safeguard for the scenario selectors; CBH approved all seven tests with these annotations. Stage 1 closed. Status rewritten for Checkpoint 5.

v0.1, 3 October 2026, twelfth entry: CBH confirmed every [CL] decision and proposal made up to Checkpoint 5; markers are kept so the origin of each decision stays visible. Stage 2 started in

a companion document (Stage 2, Process Inventory and Coupling Table v0.1). Status rewritten for Checkpoint 6.

v0.1, 3 October 2026, thirteenth entry: Stage 2 equation block 1 (E1 to E5) verified and numerically checked in the companion document. Status rewritten for Checkpoint 7.

v0.1, 3 October 2026, fourteenth entry: Stage 2 equation block 2 (E6 to E10) verified in the companion document. Ref 92 upgraded from V* to V after its DOI was retrieved (old status struck through). Status rewritten for Checkpoint 8.

v0.1, 3 October 2026, fifteenth entry: Stage 2 equation block 3 (E11 to E14) and the E8 reproduction check completed in the companion document; E14 pending its equation forms. Status rewritten for Checkpoint 9.

v0.1, 3 October 2026, sixteenth entry: Stage 2 block 4 completed (E16 verified, E14 partly verified, E15 pending coefficients); decision D10 on how accretion is computed raised for CBH. Status rewritten for Checkpoint 10.

v0.1, 3 October 2026, seventeenth entry: CBH decided D10 (own N-body accretion runs), recorded under the decisions log. Status rewritten for Checkpoint 11.

v0.1, 4 October 2026, eighteenth entry: refs 99 to 105 added for the collision-outcome model (Stage 2, accretion run design part 1), with 104 marked $\diamond$. Decision D11 on building-block iron raised in the companion document. Status rewritten for Checkpoint 12.

v0.1, 4 October 2026, nineteenth entry: CBH decided D11 (option C, two building-block iron scenarios with F3 reported for each), recorded under the decisions log. Status rewritten for Checkpoint 13.

v0.1, 4 October 2026, twentieth entry: accretion run design part 2 drafted in the companion document (hosting, benchmark B0, decision D12). Ref ± 106 added and an origin-of-life context layer proposed under open items. Status rewritten for Checkpoint 14.

v0.1, 4 October 2026, twenty-first entry: hosting decided by CBH (GitHub Actions for runs, Colab for analysis); D12 under discussion, with notes in the companion document. Status rewritten for Checkpoint 15.

v0.1, 4 October 2026, twenty-second entry: encounter check for D12 run and recorded in the companion document, with a possible units issue in the collision module flagged [CL, unconfirmed]. Status rewritten for Checkpoint 16.

v0.1, 4 October 2026, twenty-third entry: the collision module's units issue confirmed against its own paper's definition [102] (Chambers 2013 still unread); D12 still under discussion. Status rewritten for Checkpoint 17.

v0.1, 4 October 2026, twenty-fourth entry: CBH decided D12 (radius inflation settled by preregistered matched comparison P-D12; draft in the companion document, pending CBH review). Status rewritten for Checkpoint 18.

v0.1, 4 October 2026, twenty-fifth entry: CBH approved the P-D12 equivalence margins; contacting the module's author deferred by CBH. Status rewritten for Checkpoint 19.

v0.1, 4 October 2026, twenty-sixth entry: repository files for B0 drafted and tested [CL], saved to Drive pending the public repository; details in the companion document. Status rewritten for Checkpoint 20.